Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Data are 3, 7, 8, 10, 12. Find the range.

Answer: _____

P2. Find the five-number summary of 1, 3, 4, 6, 8, 9, 11. Exclude the median from each half.

Answer: _____

P3. Find the five-number summary and IQR of 1, 2, 4, 6, 8, 10, 12, 13.

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Data are 2, 4, 6, 8. Find the mean, each distance from the mean, and the MAD.

Answer: _____

P5. A box plot has five-number summary 0, 2.5, 5, 9, 12. Which interval represents the upper quarter? Does a longer interval mean more data?

Answer: _____

P6. Compare A: 4, 4, 6, 6 and B: 1, 3, 7, 9. Find each mean and MAD. Which set is more spread out?

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Data are 1, 3, 4, 5, 6, 7, 9, 11. Replace 11 with 31. Compare the old and new range and IQR.

Answer: _____

F. Data are 3, 5, 7, 9. Mean = 6. Distances are 3, __, __, 3. MAD = __.

Answer: _____

D. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

Answer: _____

Spread and box plots

Follow the slide labels. P = problem; F = your turn; D = discussion.

S. IQR measures __. MAD measures __.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. For 1, 3, 3, 5, 7, 9, 11, 13, find Q1, the median, Q3, and the IQR. Show the two halves.

Answer: _____

Name _____ Date _____ Week 9

Grade 6 Cumulative Check

30 minutes. No calculator. Show your work.

1. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

Answer: _____

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

3. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

Answer: _____

4. Find 140% of 30 points.

Answer: _____

5. 12 shells are 30% of a collection. Find the whole collection.

Answer: _____

6. How many $\frac{3}{4}$-liter groups fit in $\frac{7}{8}$ liter? Include part of a group.

Answer: _____

Grade 6 Cumulative Check

7. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

Answer: _____

8. Which is the reciprocal of $2\ \frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

Answer: _____

9. Three notebooks cost $2.45 each. What is the total cost?

Answer: _____

10. Find 9.6 ÷ 8.

Answer: _____

11. Find 1,344 ÷ 32. Verify by multiplication.

Answer: _____

12. Expand 2(x + 7).

Answer: _____

Grade 6 Cumulative Check

13. Solve $x + 9 = 23$.

Answer: _____

14. Solve $(\frac{3}{4})x = 12$.

Answer: _____

15. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

Answer: _____

16. A triangle has base 9 cm and height 4 cm. Find its area.

Answer: _____

17. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

18. Order -7, 2, -1 from least to greatest.

Answer: _____

Grade 6 Cumulative Check

19. Reflect (-3, 4) across the y-axis.

Answer: _____

20. Write an inequality for x that means at least 9. Is the boundary included?

Answer: _____

21. Find the mean of 2, 3, 7, 8.

Answer: _____

22. Find the median of 9, 1, 5, 3, 7.

Answer: _____

23. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

Answer: _____

24. Find the MAD of 1, 3, 5, 7.

Answer: _____

Data, center, and spread

The big idea

Data vary. Center describes a typical value, and spread describes the variation. A display makes both easier to see.

Words to know

statistical question: A question answered with data that are expected to vary. Example: How many books did each classmate read?

mean: The total shared equally among all data values. Example: The mean of 2 and 6 is 4.

median: The middle of ordered data. With two middles, use their mean. Example: The median of 1, 4, 8 is 4.

histogram: A display counting values in number intervals. Example: A bar can count heights from 120 to under 130 cm.

range: Largest value minus smallest value. Example: For 2, 4, 9, range is 7.

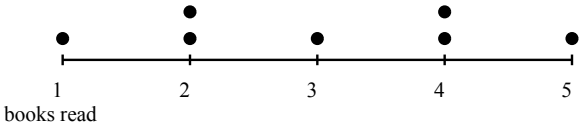
quartiles: Medians of the lower and upper halves of ordered data. Example: Exclude the middle value when the count is odd.

IQR: Upper quartile minus lower quartile. Example: Q3 = 9 and Q1 = 3 give IQR 6.

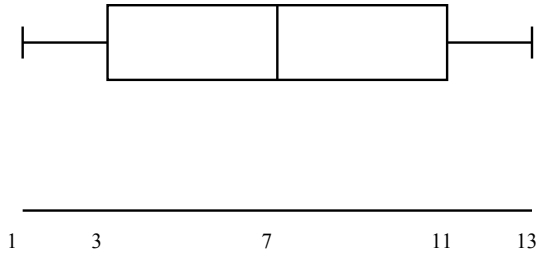
MAD: The mean of the distances from the mean. Example: Use positive distances before averaging.

How it works

The seven values balance around 3. A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.



The median splits the ordered data. Quartiles split each half. Center describes a typical value. Spread describes how far values are apart or away from the center.



Worked examples

1. Find the mean of 3, 4, 5, and 8.

The total is 3 + 4 + 5 + 8 = 20.

There are four values, so share 20 among four groups.

Mean = 20 ÷ 4 = 5. The deviations −2, −1, 0, 3 balance to zero.

Answer: 5

2. Find the IQR for 2, 4, 6, 8, 10, 12.

Split the ordered data into 2, 4, 6 and 8, 10, 12.

The two half-medians are Q1 = 4 and Q3 = 10.

IQR = 10 − 4 = 6.

Answer: 6

Common mistakes

Omits repeated values in a dot plot. Why: Treats the display as a list of categories. Fix: Cross out each raw value after plotting one dot.

Chooses one middle value in an even set. Why: Uses an odd-set rule. Fix: Mark both middle values and find their midpoint.

Uses median for every total-sharing question. Why: Does not connect mean to a fixed total. Fix: Redistribute real counters equally.

Optional preview

A 25% discount leaves 75% of the starting price. For 3(x + 2) = 18, divide both sides by 3, then subtract 2. This gives x = 4.

Try it

1. Find the mean of 2, 3, 7, 8.

2. Find the mean of 1, 4, 5, 6.

3. Find the range of 2, 5, 8, 11.

4. Find the range of 3, 4, 6, 9, 15.

5. Find the range of 0, 2, 5, 10.

Answers

1. 5 2. 4 3. 9 4. 12 5. 10

Where this connects

Mean uses equal sharing, and MAD uses distance from zero. A cumulative check links these new ideas with earlier Grade 6 work.